Fresh meat and further processing characteristics of ham muscles from finishing pigs fed ractopamine hydrochloride.

Ractopamine hydrochloride (RAC) has consistently led to an advantage in carcass cutting yields of finishing pigs and remains a common feed additive in US finishing pig diets. Less is known about the effect of RAC on further processing characteristics. Some researchers have reported advantages in ultimate pH of the LM in pigs fed RAC. If a greater ultimate pH was also observed in hams, the increased pH could affect further processing characteristics and lead to better protein interaction and improved textural properties. The objective of this experiment was to determine if RAC-fed pigs yielded hams with a greater ultimate pH, and if so, whether or not that advantage improves textural properties and water retention of further processed hams. Two hundred hams from barrows and gilts fed RAC or control diets were selected based on HCW. Hams were fabricated into 5 separate pieces to determine cutting yields, and 6 muscles were evaluated for ultimate pH. Hams were processed to make cured and smoked hams. Ractopamine increased cutting yields of the whole ham ($P < 0.0001$), inside ($P < 0.01$), outside ($P < 0.01$), and knuckle ($P < 0.01$) when expressed as a percentage of chilled side weight. Ultimate pH of the rectus femoris, vastus lateralis, and semitendinosus were all 0.06 pH units greater ($P < 0.05$), the biceps femoris was 0.04 pH units greater ($P = 0.02$), and the semimembranosus and adductor muscles were 0.03 pH units greater in pigs fed 7.4 mg/kg of RAC when compared with control pigs. Cured hams from RAC-fed pigs were heavier at all stages of production. No differences were detected in binding strengths ($P = 0.88$) or protein fat-free values ($P = 0.13$) between RAC (9.06 kg and 20.37) and control hams (9.01 kg and 20.13). Ractopamine increased cutting yields, total weight of cured hams, and ultimate muscle pH. Ractopamine can be fed to pigs to achieve the desired growth characteristic advantages and cutting yields without affecting further processed ham characteristics.